

ROYAL ISLAND, ON, Canada

WMO: 718590

Lat: 49.467N

Lon: 94.761W

Elev: 329

StdP: 97.43

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-29.5	-35.5	0.1	-31.8	-32.7	0.2	-29.3	9.3	-11.3	8.3	-13.3	2.2	270	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	28.8	21.2	27.3	20.2	25.9	19.5	22.8	26.7	21.7	25.5	20.7	24.4	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.9	25.2	20.4	15.7	24.1	19.4	14.7	22.9	69.5	26.8	65.2	25.5	61.4	24.4	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	6.9	DB	-36.1	31.7	2.8	1.7	-38.1	32.9	-39.7	33.9	-41.2	34.8	-43.2	36.1	(4)
(5)			WB	-35.9	24.7	2.9	1.6	-38.0	25.8	-39.7	26.7	-41.3	27.6	-43.4	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-15.4	-13.4	-4.8	3.8	11.6	17.6	20.9	20.1	15.2	7.0	-2.0	-11.4	(6)
(7)		DBStd	13.95	8.09	7.62	7.41	5.57	4.64	3.28	2.76	2.96	4.12	4.64	6.01	7.60	(7)
(8)		HDD10.0	3282	788	654	458	196	37	1	0	0	7	117	360	664	(8)
(9)		HDD18.3	5368	1047	887	716	437	212	51	8	17	110	351	609	923	(9)
(10)		CDD10.0	1164	0	0	1	9	87	228	339	312	162	25	1	0	(10)
(11)		CDD18.3	209	0	0	0	0	4	29	89	71	16	1	0	0	(11)
(12)		CDH23.3	1104	0	0	0	0	29	147	493	365	69	1	0	0	(12)
(13)		CDH26.7	197	0	0	0	0	2	14	97	73	10	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.0	2.9	3.0	3.3	3.3	2.7	2.4	2.5	2.9	3.6	3.8	3.1	(14)
(15)	Precipitation	PrecAvg	695	30	24	32	38	82	100	105	75	81	59	40	31	(15)
(16)		PrecMax	865	55	48	59	72	158	205	200	159	152	128	101	56	(16)
(17)		PrecMin	461	11	6	11	16	31	48	38	4	26	11	12	12	(17)
(18)		PrecStd	115	11	10	14	16	37	42	40	38	38	27	23	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	5.5	13.7	20.9	26.3	28.4	31.2	31.1	28.1	21.6	13.0	4.0	(19)
(20)			MCWB	1.4	2.8	8.8	11.2	15.9	20.0	23.4	22.7	20.5	14.9	8.9	1.9	(20)
(21)		2%	DB	0.5	2.9	9.2	17.4	23.7	26.5	29.0	28.7	25.2	18.5	9.5	1.5	(21)
(22)			MCWB	-0.3	1.3	4.9	9.6	15.3	19.0	21.9	20.9	19.1	13.5	6.6	0.3	(22)
(23)		5%	DB	-1.6	0.7	6.7	14.4	21.5	25.1	27.6	26.9	23.2	16.0	7.3	0.2	(23)
(24)			MCWB	-2.2	-0.2	3.7	7.9	14.1	18.4	20.9	19.9	17.6	12.0	5.2	-0.6	(24)
(25)		10%	DB	-3.8	-1.4	4.5	12.0	19.0	23.4	26.1	25.4	21.2	13.4	5.6	-1.2	(25)
(26)			MCWB	-4.4	-2.6	2.0	6.6	13.1	17.7	20.1	19.2	16.8	10.3	4.0	-1.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	3.6	9.1	12.2	19.5	22.8	25.0	24.3	22.0	16.3	9.5	2.1	(27)
(28)			MCDWB	2.9	5.0	13.0	19.5	23.2	26.1	28.9	28.6	26.4	18.8	12.2	3.6	(28)
(29)		2%	WB	-0.2	1.4	5.8	10.2	17.1	20.8	23.3	22.4	20.2	14.1	7.2	0.6	(29)
(30)			MCDWB	0.4	2.6	8.5	15.3	20.8	24.5	27.1	26.7	23.3	18.1	9.1	1.3	(30)
(31)		5%	WB	-2.2	-0.1	3.8	8.7	15.2	19.5	22.1	21.2	18.7	12.4	5.5	-0.6	(31)
(32)			MCDWB	-1.5	0.8	6.4	13.6	19.0	23.2	25.8	25.4	22.0	15.4	7.1	0.1	(32)
(33)		10%	WB	-4.4	-2.6	2.2	7.1	13.9	18.4	21.1	20.1	17.4	10.5	3.9	-1.9	(33)
(34)			MCDWB	-3.9	-1.6	4.1	11.7	17.9	21.9	24.7	23.7	20.4	12.9	5.5	-1.2	(34)
(35)	Mean Daily Temperature Range			MDBR	9.9	11.8	11.5	10.4	9.3	8.5	8.5	7.8	6.2	5.5	7.8	(35)
(36)		5% DB	MCDBR	9.2	10.0	11.2	13.4	12.4	10.7	10.0	10.5	10.3	10.0	7.4	7.6	(36)
(37)			MCWBR	8.4	8.1	7.7	7.3	5.7	5.2	4.7	4.6	5.1	5.7	4.8	6.5	(37)
(38)			MCDBR	9.1	8.6	10.5	12.3	9.9	9.3	9.1	9.3	9.4	9.2	6.8	7.3	(38)
(39)		5% WB	MCWBR	8.4	7.1	7.4	7.0	5.6	5.4	5.2	4.8	5.3	5.8	4.8	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.247	0.259	0.293	0.332	0.349	0.372	0.378	0.369	0.346	0.303	0.271	0.246	(40)
(41)		taud		2.326	2.306	2.377	2.434	2.455	2.411	2.420	2.465	2.507	2.590	2.485	2.349	(41)
(42)		Ebn at Noon		824	900	924	917	911	887	875	869	856	847	796	774	(42)
(43)		Edh at Noon		66	87	98	104	106	112	110	100	87	67	58	56	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.30	2.46	3.66	4.75	5.38	5.62	5.96	4.94	3.63	2.10	1.13	0.91	(44)
(45)		RadStd		0.14	0.19	0.36	0.47	0.38	0.46	0.33	0.37	0.31	0.30	0.15	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page